

1. Solve the following definite integrals

(a) $\int_3^6 x^2 + 9x \, dx$

(c) $\int_{-3}^4 \frac{x^2 + 5x + 6}{x + 2} \, dx$

(e) $\int_1^\infty \frac{1}{x^2} + \frac{2}{x^3} + \frac{3}{x^4} \, dx$

(b) $\int_0^{100} \frac{10\sqrt{x} - x}{100} \, dx$

(d) $\int_0^{2\pi} 3 \cos(x) + 2 \sin(x) \, dx$

(f) $\int_{-\infty}^\infty \frac{1}{x^2 + 1} \, dx$

2. Solve the following indefinite integrals using a u substitution

(a) $\int (4x + 3)(2x^2 + 3x) \, dx$

(c) $\int \left(-\frac{1}{x^2} + \frac{1}{2\sqrt{x}} \right) \times \sqrt{\frac{1}{x} + \sqrt{x}} \, dx$

(e) $\int \sec^2(x) (\ln(\sin x) - \ln(\cos x)) \, dx$

(b) $\int \frac{x^3 + 1}{x^4 + 4x + 4} \, dx$

(d) $\int \cos(x) \sin(\sin(x)) \, dx$

(f) $\int 16x^3 \ln(x) \, dx$

3. Solve the following indefinite integrals using integration by parts and u substitution

(a) $\int x(x + 3)^7 \, dx$

(c) $\int x e^x \, dx$

(e) $\int \cos^3(x) \, dx$

(b) $\int x^2(x + 1)^7 \, dx$

(d) $\int \sin^3(x) \, dx$

(f) $\int e^x \cos(x) \, dx$